

BST Physics Curriculum Vol 12 Ch 3 — Molecular Orbital Theory + LCAO v0.4 (Calibration #23 substance-floor refill)

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redirect)

Vol 12 Chapter 3 — Molecular Orbital Theory + LCAO

Chapter motivation

Molecular orbital (MO) theory and valence bond (VB) theory are the two foundational quantum-mechanical descriptions of molecular bonding. MO (Hund-Mulliken 1928) treats electrons as delocalized over the entire molecule; VB (Heitler-London 1927) treats electrons as localized in bond pairs. They are complementary descriptions of the same multi-electron wavefunction. Standard chemistry treatment (Atkins Molecular Quantum Mechanics, Levine Quantum Chemistry): LCAO (linear combination of atomic orbitals), bonding/antibonding MOs, hybridization ($sp/sp^2/sp^3$), π and σ bonds, frontier orbital theory (HOMO/LUMO). BST substrate-cartography reading: each construction is a substrate K-type representation operation on Bergman $H^2(D_{IV}^5)$ with $\text{Pin}(2) Z_2$ grading — LCAO is K-type linear combination, hybridization is K-type basis rotation, π/σ classification is angular momentum projection.

Section 3.0b — Reader-grade 3-level pedagogy

Level 1: MO theory = electrons delocalized; VB theory = electrons in bond pairs; both are valid substrate K-type representations on Bergman $H^2(D_{IV}^5)$. LCAO + hybridization + π/σ bonds = substrate K-type operations. Carbon's sp^3 tetrahedral hybridization ($4 = N_c + 1$) underlies organic chemistry geometric richness.

Level 2 (graduate-chemist accessible): MO theory (Hund 1927, Mulliken 1928): construct molecular orbitals as linear combinations of atomic orbitals (LCAO), $\psi_{MO} = c_A \varphi_A + c_B \varphi_B$ with coefficients determined by Roothaan-Hall equations (Hartree-Fock self-consistent field). Bonding MO (in-phase combination) lowers energy; antibonding MO (out-of-phase combination) raises energy. Electrons fill MOs per Aufbau + Pauli + Hund's rule. Valence Bond (Heitler-London

1927, Pauling 1931): electrons localized in bond pairs with covalent + ionic resonance contributions. Hybridization (Pauling 1931): atomic orbitals mix to give equivalent hybrids — sp (180° linear, acetylene $C\equiv C$), sp^2 (120° trigonal, ethylene $C=C$ + benzene aromatic), sp^3 (109.5° tetrahedral, methane CH_4). σ bonds (cylindrical symmetry along bond axis, $m_l = 0$); π bonds (nodal plane through bond axis, $|m_l| = 1$); δ bonds ($|m_l| = 2$, transition-metal rare). Frontier orbital theory (Fukui 1952): HOMO (highest occupied) + LUMO (lowest unoccupied) govern reactivity. Famous example: O_2 paramagnetism predicted only by MO theory (two unpaired electrons in degenerate π^* antibonding MOs); VB theory misses this without elaborate resonance treatment. BST substrate-cartography reading: atomic orbitals are Wallach K-type representations on D_{IV}^5 (T2438 K-type classification); LCAO is K-type linear combination producing molecular K-types on Bergman $H^2(D_{IV}^5)$; hybridization is K-type basis rotation preserving Casimir spectrum; bonding/antibonding splitting via $Pin(2)$ Z_2 grading (symmetric/antisymmetric); π/σ classification is K-type angular-momentum projection ($\sigma = m_l = 0$; $\pi = |m_l| = 1$). Carbon's sp^3 tetravalent structure (4 equivalent bonds) reads $4 = N_c + 1$ BST primary integer + identity.

Level 3 (5th-grader accessible): When atoms come together, their electron “neighborhoods” (orbitals) combine to make molecular neighborhoods. There are two ways to think about it: MO theory says electrons live everywhere in the molecule (like roaming around a whole apartment building); VB theory says electrons live mostly between specific pairs of atoms (like couples in their own apartments). Both are right — they’re two ways to describe the same thing. Carbon has a special trick called sp^3 hybridization that makes 4 equal-strength bonds pointing to the corners of a tetrahedron — this is why carbon makes all of organic chemistry. BST framework says these orbital combinations are operations on substrate “K-types” (D_{IV}^5 building blocks).

Section 3.1 — MO Theory (Hund-Mulliken)

LCAO construction: $\psi_{MO} = \sum_i c_i \varphi_i$ where φ_i are atomic orbitals. Hartree-Fock self-consistent field determines coefficients. Bonding/antibonding splitting magnitude depends on atomic orbital overlap integral $S_{AB} = \int \varphi_A^* \varphi_B d\tau$.

O_2 paramagnetism prediction: MO theory correctly predicts O_2 has two unpaired electrons in degenerate π^* antibonding orbitals $\rightarrow$ paramagnetic (sticks to magnet). VB theory misses this without elaborate resonance treatment. This was historical evidence for MO theory’s accuracy.

BST framing: LCAO = K-type linear combination on Bergman $H^2(D_{IV}^5)$; bonding/antibonding splitting = $Pin(2)$ Z_2 grading (symmetric $+1$ / antisymmetric -1 eigenvalue projections).

Section 3.2 — VB Theory + Hybridization (Heitler-London + Pauling)

VB construction: electron pair localization between two atoms with covalent ($A\uparrow B\downarrow + A\downarrow B\uparrow$) + ionic ($A^{2-}B^0 + A^0B^{2-}$) contributions. Resonance hybrid: actual structure = superposition of canonical structures (benzene Kekulé structures classic example).

Hybridization sequence: - sp (180° linear, 2 hybrids): acetylene $C\equiv C$, CO_2 - sp^2 (120° trigonal, 3 hybrids): ethylene $C=C$, benzene aromatic ring - sp^3 (109.5° tetrahedral, 4 hybrids): methane CH_4 , all sp^3 saturated organic

BST framing: hybridization is K-type basis rotation on substrate. The $4 = N_c + 1$ sp^3 structure is the geometric foundation of organic chemistry (Vol 12 Ch 8 cross-link).

Section 3.3 — π and σ Bond Classification

σ bonds: cylindrical symmetry along bond axis; angular momentum projection $m_l = 0$; head-on orbital overlap. π bonds: nodal plane through bond axis; angular momentum projection $|m_l| = 1$; side-on overlap (above + below bond axis). δ bonds: $|m_l| = 2$; rare, transition metal complexes (Re-Re quadruple bond in $[Re_2Cl_8]^{2-}$).

BST framing: $\sigma/\pi/\delta$ classification is substrate K-type angular-momentum projection at each m_l value. Sequence ($m_l = 0, \pm 1, \pm 2$) reads substrate angular momentum ladder.

Section 3.4 — Frontier Orbital Theory (HOMO/LUMO)

HOMO (highest occupied molecular orbital) + LUMO (lowest unoccupied molecular orbital) govern chemical reactivity per Fukui frontier orbital theory (1952). Reactions proceed via HOMO-LUMO interactions; Woodward-Hoffmann rules predict allowed/forbidden pericyclic reactions.

BST framing: HOMO/LUMO energies are substrate K-type Casimir eigenvalues at occupation boundary. Frontier orbital energy gaps emerge from substrate spectrum on Bergman $H^2(D_{IV}^5)$.

Section 3.5 — Connection

- Vol 12 Ch 2 Chemical Bonding from Substrate (bond class framework)
- Vol 5 QM K-type representation theory
- Vol 3 Ch 7 Atomic orbital sequence ($2l+1$) reading
- Vol 12 Ch 4 Chemical Reactions + Thermo (reactivity follows MO theory)
- Vol 12 Ch 5 Reaction Kinetics + Transition States (HOMO/LUMO reactivity)
- Vol 12 Ch 8 Organic Carbon (sp^3 geometric chemistry full treatment)

Honest scope: K-type representation framework provides organizing principle; specific quantitative MO calculations from substrate Hamiltonian remain multi-year SP-31 program. Hybridization integer-count $4 = N_c + 1$ is I-tier framework integer-match.

— Lyra, Vol 12 Ch 3 v0.4 chapter-grade narrative REFILLED per Cal #104 + Calibration #23, Saturday 2026-05-23 EDT